WORKED SOLUTION

STEP III 2020, Question 11

Probability and Statistics • Functions of Random Variables

STEP III 2020, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

A strictly decreasing function reverses inequalities, and because f is its own inverse we can apply it to both sides of $\mathrm{f}(X) \leq y$ to get an event about X alone. The expectation then comes from integrating by parts, which swaps f' for f. In part (ii) the relation between Z and X is symmetric in the two variables, so $Z = \mathrm{f}(X)$ for a self-inverse f and part (i) applies. The inequality is then just the statement that a variance is positive.

Solution

Part (i)

Since f^{-1} is defined on all of $[a, b]$, f is a bijection of $[a, b]$ onto itself, so Y takes values in $[a, b]$. Fix $y \in [a, b]$. Because f is strictly decreasing, applying it to both sides of an inequality reverses the inequality, and $\mathrm{f}(\mathrm{f}(x)) = x$. Hence, for $x \in [a, b]$,

$$\mathrm{f}(x) \leq y \iff \mathrm{f}(\mathrm{f}(x)) \geq \mathrm{f}(y) \iff x \geq \mathrm{f}(y).$$

(The reverse implication holds in the same way, by applying f to $x \geq \mathrm{f}(y)$.) Since X is uniform on $[a, b]$ and $\mathrm{f}(y) \in [a, b]$,

$$\mathrm{P}(Y \leq y) = \mathrm{P}(X \geq \mathrm{f}(y)) = \frac{b - \mathrm{f}(y)}{b - a},$$

as required.

Differentiating, the probability density function of Y is

$$g(y) = -\frac{\mathrm{f}'(y)}{b - a} \quad \text{for } a \leq y \leq b,$$

and zero otherwise. This is non-negative, since f is decreasing. Then, integrating by

parts with $u = y^2$ and $\mathrm{d}v = -\mathrm{f}'(y)\,\mathrm{d}y$,

$$\begin{aligned}\mathrm{E}(Y^2) &= \frac{1}{b-a}\int_a^b y^2(-\mathrm{f}'(y))\,\mathrm{d}y \\ &= \frac{1}{b-a}\Big[-y^2\mathrm{f}(y)\Big]_a^b + \int_a^b \frac{2y\,\mathrm{f}(y)}{b-a}\,\mathrm{d}y \\ &= \frac{-b^2a + a^2b}{b-a} + \int_a^b \frac{2y\,\mathrm{f}(y)}{b-a}\,\mathrm{d}y,\end{aligned}$$

using $\mathrm{f}(b) = a$ and $\mathrm{f}(a) = b$. The first term is $-ab(b-a)/(b-a) = -ab$. Renaming the dummy variable y as x gives

$$\mathrm{E}(Y^2) = -ab + \int_a^b \frac{2x\,\mathrm{f}(x)}{b-a}\,\mathrm{d}x.$$

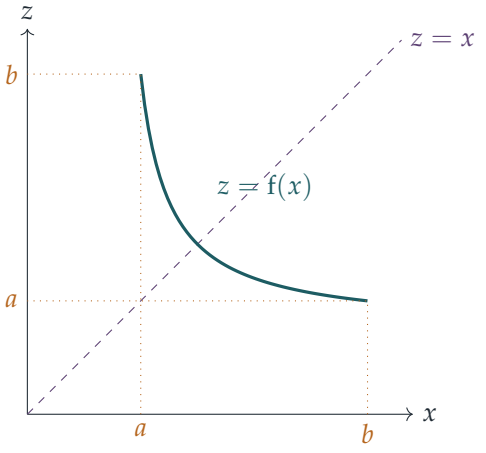
Part (ii)

Note that $\frac{1}{c} = \frac{1}{a} + \frac{1}{b}$ gives $c = \dfrac{ab}{a+b}$, and $c < a$ because $\frac{1}{c} > \frac{1}{a}$. Solving for Z,

$$\frac{1}{Z} = \frac{1}{c} - \frac{1}{X} = \frac{X-c}{cX}, \qquad \text{so} \qquad Z = \mathrm{f}(X) \ \text{ where } \ \mathrm{f}(x) = \frac{cx}{x-c}.$$

Since $x \geq a > c$, the denominator is positive on $[a,b]$. We check that f satisfies the three conditions of part (i).

- $\frac{1}{\mathrm{f}(a)} = \frac{1}{c} - \frac{1}{a} = \frac{1}{b}$, so $\mathrm{f}(a) = b$. Similarly $\frac{1}{\mathrm{f}(b)} = \frac{1}{c} - \frac{1}{b} = \frac{1}{a}$, so $\mathrm{f}(b) = a$.
- $\frac{1}{\mathrm{f}(x)} = \frac{1}{c} - \frac{1}{x}$ is positive and strictly increasing on $[a,b]$, so $\mathrm{f}(x)$ is strictly decreasing there.
- The relation $\frac{1}{z} + \frac{1}{x} = \frac{1}{c}$ is symmetric in x and z. So $z = \mathrm{f}(x)$ exactly when $x = \mathrm{f}(z)$, which means $\mathrm{f} = \mathrm{f}^{-1}$.



The graph of f runs from (a,b) to (b,a) and is symmetric in the line $z = x$, as a self-inverse function must be.

Ask yourself Why bother finding $\mathrm{E}(Z)$ as well as $\mathrm{E}(Z^2)$?

The inequality has to come from $\mathrm{Var}(Z) > 0$, and the variance needs both moments. The logarithm will appear in each moment, and its first power will cancel, leaving its square.

For both integrals, divide out:

$$\mathrm{f}(x) = c + \frac{c^2}{x-c}, \qquad x\,\mathrm{f}(x) = \frac{cx^2}{x-c} = c\Big(x + c + \frac{c^2}{x-c}\Big).$$

Write $L = \ln\Big(\dfrac{b-c}{a-c}\Big)$, which is positive because $b - c > a - c > 0$. Then

$$\mathrm{E}(Z) = \frac{1}{b-a}\int_a^b \Big(c + \frac{c^2}{x-c}\Big)\,\mathrm{d}x = c + \frac{c^2 L}{b-a}.$$

By part (i),

$$\begin{aligned}\mathrm{E}(Z^2) &= -ab + \frac{2c}{b-a}\int_a^b \Big(x + c + \frac{c^2}{x-c}\Big)\,\mathrm{d}x \\ &= -ab + \frac{2c}{b-a}\Big(\frac{b^2-a^2}{2} + c(b-a) + c^2 L\Big) \\ &= -ab + c(a+b) + 2c^2 + \frac{2c^3 L}{b-a}.\end{aligned}$$

Now $c(a+b) = ab$, so the first two terms cancel. Subtracting $\mathrm{E}(Z)^2 = c^2 + \dfrac{2c^3 L}{b-a} + \dfrac{c^4 L^2}{(b-a)^2}$, the terms in L cancel and we get

$$\mathrm{Var}(Z) = c^2 - \frac{c^4 L^2}{(b-a)^2}.$$

Z is not constant: by part (i) its distribution function $\big(b - \mathrm{f}(y)\big)/(b-a)$ increases strictly from 0 to 1 across $[a, b]$. So $\mathrm{Var}(Z) > 0$, which gives

$$L^2 < \frac{(b-a)^2}{c^2}.$$

Both L and $(b-a)/c$ are positive, so we may take square roots:

$$\ln\Big(\frac{b-c}{a-c}\Big) < \frac{b-a}{c}.$$

Checks

- In part (i), try the simplest self-inverse example $\mathrm{f}(x) = a + b - x$. Then Y is uniform on $[a, b]$, and the formula gives $\mathrm{E}(Y^2) = -ab + \frac{2}{b-a}\int_a^b x(a+b-x)\,\mathrm{d}x = \frac{1}{3}(a^2 + ab + b^2)$, which is the correct second moment of a uniform variable.

- With $a = 1$, $b = 3$ we have $c = \frac{3}{4}$ and $L = \ln 9 \approx 2.197$, while $(b-a)/c = \frac{8}{3} \approx 2.667$. The inequality holds, and $\mathrm{Var}(Z) = \frac{9}{16} - \frac{81}{256} \cdot \frac{(\ln 9)^2}{4} \approx 0.180$. Integrating f and f^2 numerically by computer gives the same variance.
- As $b \to a$, both sides of the inequality tend to 0, and $\mathrm{Var}(Z) \to 0$, as it should: Z is squeezed into $[a, b]$.

Common slips

- Forgetting that a decreasing function reverses the inequality, and so writing $\mathrm{P}(Y \leq y) = \mathrm{P}(X \leq \mathrm{f}(y))$.
- Leaving out the minus sign in the density $-\mathrm{f}'(y)/(b-a)$. A density cannot be negative.
- Not saying why $\mathrm{Var}(Z)$ is *strictly* positive, or taking square roots without noting that both sides are positive.

Beyond STEP

Part (i) is a special case of the transformation rule for densities: if $Y = \mathrm{f}(X)$ with f monotonic, then $g_Y(y) = g_X\big(\mathrm{f}^{-1}(y)\big)\,\big|(\mathrm{f}^{-1})'(y)\big|$. Getting a strict inequality from $\mathrm{Var}(Z) > 0$ is an example of a powerful trick: $\mathrm{E}(Z)^2 \leq \mathrm{E}(Z^2)$ is Jensen's inequality (or Cauchy–Schwarz) in disguise, and choosing a clever random variable proves many classical inequalities. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

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